Fiscal Year:	FY 2012	**Task Last Updated:**	FY 11/13/2012
PI Name:	Nielsen, Sheila Ph.D.		
Project Title:	The fungal response to simulated microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Comments:	PI name change to Sheila Nielsen in 2014 (formerly Sheila Nielsen-Preiss)--Ed., 1/12/2015		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	07/01/2009	**End Date:**	06/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: Received extension to 6/30/2012 per A. Chu/ARC (Ed., 4/29/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hyman, Linda (Boston University)		
Grant/Contract No.:	NNX09AM79G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Modeling the microgravity environment, using rotating suspension culture bioreactors, allows investigation of eukaryotic cellular responses in a ground-based model and provides insights into the impact of space flight on cellular physiology. We have previously demonstrated that simulated microgravity induces morphologic changes and differential expression of several groups of yeast genes, including those related to morphogenic transformation, polarity determination, and the unique environmental stress of microgravity. The focus of this application is to characterize the molecular mechanisms responsible for the eukaryotic microbial response to microgravity using ground-based models. Choosing organisms from the fungal world is based on two criteria; (1) the microbial system is a powerful model to examine fundamental aspects of cell biology common to eukaryotic systems and, (2) infection by yeast pathogens (Candida albicans) is a concern for human health. The overall objective of this proposal is to examine the hypothesis that cells exposed to microgravity undergo phenotypic and genotypic changes that alter polarity determination. The net result of altered polarity is demonstrated by filamentation in Candida albicans, a physiological feature consistent with increased pathogenicity. Signal transduction pathways involved in sensing mechanoreception driving these responses will be addressed. Ultimately, the propensity for increased biofilm formation and virulence will also be evaluated in Candida albicans subjected to simulated microgravity. Given the flexible requirements for yeast growth and the availability of molecular tools in the yeast system, these ground-based studies can be readily adapted to space-flight testing.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Using the Candida albicans yeast system, we have demonstrated that simulated microgravity affects differential expression of several groups of genes, including those related to morphogenic transformation, environmental sensing and ergosterol biosynthesis; induces morphologic changes such as filamentation, biofilm formation, and colony morphology; and results in an increased resistance to the antimicrobial agent Amphotericin B. The constellation of these physiological responses is consistent with a potential increase in the virulence of this opportunistic yeast pathogen. The data obtained in these studies provides a foundation on which to explore microgravity- and other extreme environment-induced physiological changes in a eukaryotic pathogen and further establishes LSMMG as a model system.
Task Progress:	The overall objective of this proposal was to examine the hypothesis that cells exposed to microgravity undergo physical changes that affect the ability to sense direction, or have an altered polarity. One result of altered polarity in yeast is a changed budding pattern in Saccharomyces cerevisiae (Brewer's yeast) and filamentation in Candida albicans (an opportunistic pathogen) consistent with increased pathogenicity. Although our initial proposal included studies on S. cerevisiae and C. albicans, the C. albicans work became a priority based on interesting research results and the potential implications of the results as an opportunistic yeast pathogen. Ultimately, phenotypic markers associated with increased virulence in C. albicans, and the environmental cues that trigger them, were evaluated following exposure to low-shear modeled microgravity. The studies performed in this proposal were designed to be readily replicated in spaceflight conditions. Indeed, our proposal for a flight opportunity was approved and C. albicans was flown as Micro-6 on SpX CRS-1. The conclusion of this grant included a transition to the flight studies, which will be reported under separate cover. Candida albicans is an opportunistic fungal pathogen responsible for a variety of cutaneous and systemic human infections. Virulence of C. albicans increases upon exposure to some environmental stresses; therefore, we explored phenotypic responses of C. albicans following exposure to the environmental stress of low-shear modeled microgravity (LSMMG). Upon long-term (12 day) exposure to low-shear modeled microgravity, C. albicans transitioned from yeast to filamentous forms at a higher rate than observed under control conditions. Consistently, genes associated with cellular morphology, including Yeast Wall Protein (YWP1) and Hyphal Wall Protein (HWP1) were differentially expressed in a time-dependent manner. Biofilm communities, credited with enhanced resistance to environmental stress, formed in the modeled microgravity bioreactor and had a more complex structure than those formed in control conditions. Likewise, the expression of genes associated with biofilm formation and the yeast to hyphal transition, including adhesin genes (ALS1 and ALS3) and the transcriptional regulator, BCR1, was altered following exposure to LSMMG. In addition to the LSMMG-induced alterations in community structure observed with biofilm formation, cells exposed to low-shear modeled microgravity displayed phenotypic switching, observed as a near complete transition from smooth to "hyper" irregular wrinkle colony morphology. This colony phenotype has been associated with cells that also demonstrate an increased aggressiveness in terms of causing infection. Consistent with the presence of biofilm communities and increased rates of phenotypic switching, cells exposed to modeled microgravity were significantly more resistant to the antifungal agent Amphotericin B. Together, these data indicate that C. albicans adapts to the environmental stress of low-shear modeled microgravity by demonstrating virulence-associated phenotypes. C. albicans appears to maintain a semi-stable phenotypic state while retaining the flexibility necessary for rapid adaptation to environmental stresses, including filamentation, biofilm formation, phenotypic switching, and changes in the expression of several hundred genes. Data presented herein support a time-dependent, LSMMG-induced increase in cellular filamentation, altered colony morphology, biofilm formation and maturation, and antifungal resistance of C. albicans. Individually, and certainly in concert, these phenotypic transitions are suggestive of organisms that also have an increased propensity to cause infection and evade therapy. Further studies indicated that the yeast phenotypic adaptations to the low fluid shear environment were, at least in part, modulated by increased CO2 levels in the microenvironment. LSMMG appears to increase the potential for pathogenicity of C. albicans and, thus, may also serve as a model system for analyzing the low-shear in vivo environment often associated with C. albicans infections. The possibility that environmental sensing of the microgravity/low fluid-shear environment triggers a transition to a more pathogenic form of the organism could have clear implications for human health. The host-microbe interactions in spaceflight are important in their own right, but can also be extrapolated to similar interactions associated with the common, ground-based population. The immune system becomes generally less resilient upon aging. Furthermore, the number of immunocompromised individuals is increasing as a result of the use of chemotherapy and other immunosuppressant drugs. Analysis of the effects of LSMMG on pathogen virulence could provide an informed foundation to identify astronaut risk, as well as provide better understanding of the more generally applicable molecular mechanisms involved in host-pathogen interactions. A manuscript has been submitted to PLos One and is in review (see Bibliography below); another is in preparation for submission (Searles SC, Hyman LE, Nielsen-Preiss SM "Effects of modeled microgravity on C. albicans ergosterol").

Bibliography Type:	Description: (Last Updated: 06/23/2023)
Articles in Peer-reviewed Journals	Searles SC, Hyman LE, Nielsen-Preiss SM. "Physiological responses of Candida albicans to modeled microgravity are mediated by altered CO2 concentration." PLoS One. Submitted and in review as of November 2012. , Nov-2012